

## IV

### The Gloomy Presentiments of Parson Malthus and David Ricardo

**I**n addition to the omnipresent problem of poverty, a bothersome question worried England throughout most of the eighteenth century: the question of how many Englishmen there were. The worrisome aspect of the problem lay in the fact that England's natural enemies on the Continent bulged with what, to British eyes, must have appeared as a veritable flood of humanity, while England, with her slender resources, was convinced that her own population was on the decline.

Not that England was any too sure exactly how many Britishers there were; in hypochondriacal fashion, she preferred to worry in a factual vacuum. It would not be until 1801 that the first real census would be held and when it came it would be heralded as totally subversive of the last remnants of English liberty. Hence, Britain's earlier knowledge about the state of her human resources depended on the efforts of amateur statisticians: Dr. Price, a Nonconformist minister; Houghton, an apothecary and dealer in coffee and tea; and Gregory King, by trade a maker of maps. ,

Drawing on records of hearth taxes and baptismal registers, in 1696 King reckoned up the number of souls in the British Isles as something near five and a half

million—what seems to have been an extraordinarily accurate estimate. But King was concerned not only with the contemporary state of affairs. Looking into the future he wrote: “In all probability, the next doubling of the people of England will be in about six hundred years to come, or by the year of our Lord 2300. . . . The next doubling after that will be, in all probability, in less than twelve or thirteen hundred years, or by the year of our Lord 3500 or 3600. At which time the kingdom will have 22 million souls ... in case,” the map maker added circumspectly, “the world should last so long.”

But by the time of Adam Smith, King’s projection of a gently rising population had given way to another view. Comparing eighteenth-century records of hearth-money taxes with those of an earlier day. Dr. Richard Price proved conclusively that the population of England had declined by over thirty per cent since the Restoration. The validity of his computation was obviously dubious and other investigators hotly disputed his findings; nevertheless what Dr. Price believed to be so was taken as fact, and what with the political exigencies of the times, highly unpalatable fact. “The decay of population,” moaned William Paley, the theologian reformer, “is the greatest evil the state can suffer, and the improvement of it the object which ought ... to be aimed at, in preference to every other political purpose whatsoever.” Paley was not alone in his belief; the younger Pitt, the Prime Minister, even introduced a new poor-relief bill for the specific purpose of boosting the population. The bill was to pay liberal allowances for children, since it was quite apparent to Pitt that by having children a man enriched his country,” even if his offspring should turn out to be paupers.

What is striking about the population question to our modern eyes is not whether England actually was or was not in danger of petering out as a nation. In retrospect, what is interesting is how harmonious either view of the population problem was with a philosophy which put its

laH^1 ^ reaS^n’ and ProSress- Was the population declining? Then it should be encouraged to grow.

and grow it would under the august auspices of the laws which Adam Smith had shown to be the guiding principles of a free-market economy. Was the population growing? All to the good, since everyone agreed that a growing population was a national source of wealth. No matter which way one cut the cake, the result was favorable to an optimistic prognosis for society; or to put it differently, there was nothing in the population question, as it was understood, to shake men's faith in their future!

Perhaps no one summed up this optimistic outlook so naively and completely as William Godwin. Godwin, a minister and pamphleteer, looked at the vulgar world about him and shrank back in dismay. But he looked into the future and what he saw was good. In 1793 he published *Political Justice*, a book which excoriated the present but which gave promise of a distant future in which "there would no longer be a handful of rich and a multitude of poor. . . . There will be no war, no crime, no administration of justice, as it is called, and no government. Besides this there will be no disease, anguish, melancholy, or resentment." What a wonderful vision! It was, of course, highly subversive, for Godwin's Utopia called for complete equality and for the most thoroughgoing anarchic communism: even the property contract of marriage would be abolished. But in view of the high price of the book (it sold for three guineas) the Privy Council decided not to prosecute the author, and it became the fashion of the day in the aristocratic salons to discuss Mr. Godwin's daring ideas.

One home in which this debate took place was Albury House, not far from Guildford, where there resided a curious old gentleman who was described by *Gentlemen's Magazine* on his death in 1800 as "an eccentric character in the strictest sense of the term." This eccentric was Daniel Malthus, a friend of David Hume and a passionate admirer of Rousseau with whom he had gone on local botanizing walks and from whom he had received a herbarium and a set of books in one of the French philosopher's recurrent urges for self-dispossession. Like so many leisurely but inquiring gentlemen of his day, Dan-

iel Malthus enjoyed nothing better than a stimulating intellectual dialogue, and for an opponent he usually turned to his gifted son, the Reverend Thomas Robert Malthus.

Quite naturally Godwin's paradise came up for consideration and as might perhaps be expected of an eccentric disciple of Rousseau, Malthus the elder felt sympathetically inclined toward the supremely rational Utopia. But young Malthus was not so hopeful as his father. In fact, as the argument progressed, he began to see an insurmountable barrier between human society as it existed and this lovely imaginary land of everlasting peace and plenty. To convince his father he wrote his objections down at length, and so impressed was Daniel Malthus with his son's ideas that he suggested the thesis be published and presented to the public.

It was. In 1798 an anonymous treatise of fifty thousand words appeared on the scene. It was entitled *An Essay on the Principle of Population as It Affects the Future Improvement of Society*, and with it perished at one blow all the fond hopes of a harmonious universe. In a few pages young Malthus pulled the carpet from under the feet of the complacent thinkers of the times, and what he offered them in place of progress was a prospect meager, dreary, and chilling.

For what the essay on population said was that there was a tendency in nature for population to outstrip all possible means of subsistence. Far from ascending to an ever higher level, society was caught in a hopeless trap in which the human reproductive urge would inevitably shove humanity to the sheer brink of the precipice of existence. Rather than headed for Utopia, the human lot was forever condemned to a losing struggle between ravenous and multiplying mouths and the eternally insufficient stock of Nature's cupboard, however diligently that cupboard might be searched.

No wonder that after he read Malthus, Carlyle called economics the dismal science," and that poor Godwin complained that Malthus had converted friends of progress into reactionaries by the hundreds.

In one staggering intellectual blow Malthus undid all the roseate hopes of an age oriented toward self-satisfaction and a comfortable vista of progress. But as if this were not enough, at the same time a quite different kind of thinker was also preparing the coup d'état for yet another of the lulling assumptions of the late eighteenth and early nineteenth centuries. David Ricardo, an astonishingly successful stockbroker, was soon to outline a theory of economics which, while less spectacular than Malthus' inundation of humanity, would be in its own quiet way just as devastating to the pleasant premises of the age of Adam Smith.

For what Ricardo foresaw was the end of a theory of society in which everyone moved together up the escalator of progress sketched out by Smith. On the contrary, Ricardo saw that the escalator worked with different effects on different classes, that some rode triumphantly to the top, while others were carried up a few steps and then kicked back down to the bottom. Worse yet, those who kept the escalator moving were not those who rose with its motion, and those who got the full benefit of the ride did nothing to earn their reward. And to carry the metaphor one step further, if you looked carefully at those who were ascending to the top, you could see that all was not well here either; there was a furious struggle going on for a secure place on the stairs.

Society to Adam Smith was a great family; to Ricardo it was a bitter contest for supremacy, and small wonder that he should have seen it as such. In the forty years since the *Wealth of Nations* England had divided into two hostile camps: the rising industrialists, busy with their factories and fighting for parliamentary representation and social prestige, and the great landowners, a rich powerful, and entrenched aristocracy, who looked resentfully at the encroachments of the brassy nouveaux

It was not the fact that the capitalists were making money which enraged the landowners. It was the undeniable fact that they kept insisting that fees were too high. For what had happened in the short space

time since Adam Smith was that England, which had long been a grain-exporting nation, was now being forced to buy foodstuffs from abroad. Despite the mutterings of Dr. Price who saw England's population rapidly dwindling away, the actual growth of population had caused the demand for grain to exceed the supply and had quadrupled the price of a bushel of wheat. And as prices rose, so did profits: on a farm in East Lothian, Scotland, profits and rent together averaged fifty-six per cent of invested capital; on another farm of three hundred acres owned by a Mr. Birkhead—a very typical medium-sized establishment—profits were £88 in 1790, £121 in 1803, and £160 ten years later. And on the large estates of thousands of acres profits rose accordingly.

As grain soared, enterprising merchants began to buy wheat and corn abroad and bring them into the country. Quite naturally, the landlord looked on this practice with dismay. Fanning was not merely a way of life for the aristocracy, it was a business—a big business. On the Reevesby estate in Lincolnshire in 1799, for instance, Sir Joshua Banks needed two rooms for his offices, separated them with a fireproof wall and an iron door, and was proud of the fact that it took a hundred and fifty-six drawers to classify all the papers pertaining to the farm. Although such a landlord lived on his land and loved the land, although he saw his tenants daily, joined societies for the purpose of discussing crop rotation and the virtues of competing fertilizers, he did not lose sight of the fact that his income depended on the price at which he sold his crop.

Hence the flow of inexpensive grain from overseas was hardly viewed in a tolerant light. But fortunately for the landlord, the means were readily at hand to combat this distressing development. Dominating Parliament, the landlord simply legislated himself an ironclad system of

Passes and Corn Laws, which imposed sliding duties on the importation of grain; the lower the domestic price fell, the higher went the duty. In effect a floor was established to keep low-priced wheat permanently out of the English market.

But by 1813 the situation had gotten out of hand. Bad crops and the war with Napoleon conspired to bring about virtual famine prices. Wheat sold at a price of 118 shillings a quarter—approximately 14 shillings per bushel. Thus a bushel of wheat sold for a price equal to nearly twice a workman's whole weekly wage—to put this into perspective, compare the highest price ever reached by American wheat: \$3.50 per bushel in 1920 when weekly wages averaged \$26.00.

Patently, the price of grain was fantastic, and what to do about it became a question of enormous moment to the country. Parliament studied the situation carefully—and came up with the solution that the duty on foreign grain should be raised still higher! The rationale was that higher prices in the short run would act to stimulate a larger production of English wheat in the long run.

This was too much for the industrialists to take. Contrary to the landed proprietors, the capitalists wanted cheap grain, for the price of food largely determined the amount they would have to pay for labor. It was not out of humanitarian motives that the industrialists fought for cheaper food. A great London banker, Alexander Baring, declared in Parliament: . . . the labourer has no interest in this question; whether the price be 84 shillings or 105 shillings a quarter, he will get dry bread in the one case and dry bread in the other." By this Baring meant that regardless of the price of bread, the laborer would get wages enough to buy his crust and no more. But from the point of view of those who met payrolls and sought after profits, it made a vast deal of difference whether grain—and wages—were cheap or dear.

The business interests organized; Parliament found itself flooded with more petitions than it had ever received before. In view of the temper of the country, it became obviously inexpedient to push through the new higher Corn Laws without some deliberation. New committees were appointed in Commons and in Lords, and the issue was temporarily shelved. Fortunately, the next year saw the defeat of Napoleon and grain prices subsided again

toward more normal levels. But it is an index to the political power of the landholding class that thirty years would have to pass until the Corn Laws were finally wiped from the books and cheap grain permitted to come freely into Britain. . .

Writing in the midst of such a period or crisis, it is not difficult to understand why David Ricardo saw economics in a different and far more pessimistic light than Adam Smith. Smith had looked at the world and had seen in it a great concert; Ricardo saw a vicious conflict. To the author of the *Wealth of Nations* there was every reason to believe that everyone could share in the benefits of a benign providence; to the inquiring stockbroker writing about a half century later, society was not only rent into warring groups, but it seemed inescapably true that the rightful winner of the conflict—the hard-working industrialist—was bound to lose. For Ricardo believed that the only class that would benefit from the progress of society was the landlord—unless his hold on the price of grain could be broken.

"The interest of the landlords is always opposed to the interest of every other class in the community/' he wrote in 1815, and with that unequivocal sentence an undeclared war became recognized civil strife. And with the open declaration of hostilities there perished the last forlorn hope that this might after all turn out to be the best of all possible worlds. Now it seemed that if society did not drown in the Malthusian swamp of humanity, it would tear itself to pieces in the fight to secure places on David Ricardo's treacherous moving stairs

We must look more closely at the profoundly disturbing ideas of the gloomy parson and the skeptical stockbroker. But first let us look at the men themselves.

It would be hard to imagine two persons more widely separated in their backgrounds and careers than Thomas Robert Malthus and David Ricardo. Malthus, as we know, was the son of an eccentric member of the English upper middle class; Ricardo was the son of a Jewish

™ ei;c,hant-broker who had immigrated from Holland. Malthus was tenderly tutored for a university under the guidance of a philosophically minded father (one of his tutors went to jail for expressing the wish that the French revolutionaries would invade and conquer England)-Ricardo went to work for his father at the age of fourteen. Malthus spent his life in academic research; he was the first professional economist, teaching at the college founded in Haileybury by the East India Company to train its young administrators; Ricardo set up in business for himself at the age of twenty-two. Malthus was never well-to-do; by the time he was twenty-six, Ricardo who had started with a capital of eight hundred pounds—was financially independent, and in 1814, at the age of forty-two, he retired with a fortune variously estimated to be worth between £500,000 and £1,600,000.

Yet oddly enough it was Malthus, the academician, who was interested in the facts of the real world, and Ricardo, the man of affairs, who was the theoretician; the businessman cared only for invisible “laws” and the professor worried whether these laws fitted the world before his eyes. And as a final contradiction, it was Malthus with his modest income who defended the wealthy landowner, and Ricardo, a man of wealth and later a landlord himself, who fought against their interests.

Different as they were in background, training, and career, so they were accorded utterly different receptions. As for poor Malthus, in the words of a biographer, James Bonar: “He was the best abused man of his age. Bonaparte himself was not a greater enemy of his species. Here was a man who defended small-pox, slavery, and child-murder—a man who denounced soup-kitchens, early marriages, and parish allowances—a man who ‘had the impudence to many after preaching against the evils of a family/ ” “From the first,” says Bonar, “Malthus was not ignored. For thirty years it rained refutations.”

Such abuse was bound to befall a man who urged “moral restraint” on the world. And yet Malthus was neither a prude (by the standards of his times) nor,

certainly an ogre. It is true that he urged the abolition of poor relief and even opposed housing projects for the working classes. But all this was done with the sincerest interest of the poorer classes at heart—and indeed may be contrasted with the view of some contemporary social theorists who suggested blandly that the poor be allowed to die peacefully in the streets.

Hence Malthus' position was not so much a hard-hearted as a supremely logical one. Since according to his theory the basic trouble with the world was that there were too many people in it, anything which tended to promote "early connections" only aggravated the sum of mankind's misery. A man for whom "at Nature's mighty feast there is no cover" might be kept alive by charity, but since he would then propagate, such charity was only cruelty in disguise.

But logic does not always win popularity, and a man who points out the gloomy end of society can hardly expect to gain popular esteem. No doctrine was ever so reviled: Godwin described his theory as "that black and terrible demon that is always ready to stifle the hopes of humanity." In the eyes of less sophisticated readers, the demon was not so much Malthus' theory as the figure of the Reverend himself.

Ricardo, on the other hand, was a man on whom Fortune smiled from the start. A Jew by birth, he had broken with his family and adopted Quakerism to marry a handsome Quaker girl with whom he had fallen in love; but in a day when tolerance was hardly the rule—his father had traded in a part of the Exchange known as the Jews Walk—Ricardo achieved both social status and widespread private respect. Later in life, when he was in Commons, he was called on to speak from both sides of the House. "I have no hope," he said, "of conquering the alarm with which I am assailed the moment I hear the sound of my own voice." That voice was described by one witness as "harsh and scream-y," by another as "sweet and pleasant" although "pitched extremely high"; but when it spoke, the House listened. With his earnest and brilliant expositions which ignored the toss of events

and concentrated on the basic structure of society “as if he had dropped from another planet,” Ricardo became known as the man who educated Commons. Even his radicalism—he was a strong supporter of freedom of speech and assembly, and an opponent of parliamentary corruption and Catholic persecution—did not detract from the veneration in which he was held.

It is doubtful whether his admirers understood much of what they read, for there is no more difficult economist to understand than Ricardo. But although the text might have been complex and involved, its import was plain: the interests of the capitalists and the landlords were irrevocably opposed and the interests of the landlords were inimical to the community. Hence, whether they understood him or not, the industrialists made him their champion: political economy even became so popular with them that ladies who hired governesses inquired whether they could teach its principles to their children.

But while Ricardo, the economist, walked like a god (although he was a most modest and retiring person), Malthus was relegated to a lower status. His essay on population was read, admired, and then disproved again and again—the very vehemence of the disproofs a disquieting testimony to the strength of his thesis. And while Ricardo’s ideas were avidly discussed, Malthus’ contributions to economics—aside from his essay on population—were largely looked on with a kind of benevolent tolerance, or ignored. For Malthus had a sense that all was not well with the world, but he was utterly incapable of presenting his arguments in a clear-cut logical fashion: he was even heretical enough to suggest that depressions—“general gluts,” he called them—might upset society, an idea which Ricardo had no trouble in proving to be absurd. How exasperating for a modern reader! Intuitive and fact-minded, Malthus had a nose for trouble, but his woolly-headed expositions had no chance against the incisive brilliance of the stockbroker who saw the world only as a great abstract mechanism.

Hence they argued about everything. When Malthus published his own

1820 Ricardo went to the trouble of taking some 220-odd pages of notes to point out the flaws in the Reverend's arguments, and Malthus positively went out of his way in his book to expose the fallacies he was sure were inherent in Ricardo's point of view.

Strangest of all, the two were the closest of friends. They met in 1809 after Ricardo had published a series of masterful letters to the *Morning Chronicle* on the question of the price of bullion, and then had demolished a Mr. Bosanquet, who was rash enough to venture an opposing view. First James Mill and then Malthus sought out the author of the letters and a friendship formed among all three which endured to the end of their lives. A stream of correspondence passed between them, and they visited each other endlessly. "They hunted together in search of the Truth," wrote Maria Edgeworth, a contemporary writer, in a charming diary, and huzzaed when they found her, without caring who found her first."

It was not all serious discussion; these were very human beings. Malthus, whether out of deference to his theories or other reasons, had married late, but he was fond of social gatherings. After his death, someone who had known him mused on his life at East India College: "The subdued jests and external homage and occasional insurrections of the young men; the archery of the young ladies; the curious politeness of the Persian professor . . . and the somewhat old-fashioned courtesies of the summer evening parties ^are all over now."

The pamphleteers compared him with Satan, but Malthus was a tall and handsome man and a gentle soul: his students called him "Pop" behind his back. He had one odd defect: from his great-great-grandfather he had inherited a cleft palate and his speech was difficult to understand; I was his worst letter and there is an amusing account of his saying into the ear trumpet of a deaf and famous lady: "Would not you like to have a look at the lakes of Killamey?" This defect and the indissoluble association of his name with overpopulation led one acquaintance to write:

Philosopher Malthus was here last week. I got an agreeable party for him of unmarried people ... he is a good-natured man and, if there are no signs of approaching fertility, is civil to every lady. . . . Malthus is a real moral philosopher, and I would almost consent to speak as inarticulately, if I could think and act as wisely.

Ricardo at home also loved to entertain: his breakfasts were famous, and he seems to have indulged in a fondness for charades. In her *Life and Letters*, Miss Edgeworth tells of one round:

coxcomb— Mr. Smith, Mr. Ricardo, Fanny, Harriet, and Maria, crowing. Ditto, ditto, combing hair. Mr. Ricardo, solus strutting, a coxcomb, very droll.

He was extraordinarily gifted as a businessman. “The talent for obtaining wealth,” wrote his brother, “is not held in much estimation, but perhaps in nothing did Mr. R. more evince his extraordinary powers than he did in business. His complete knowledge of all its intricacies—his surprising quickness at figures and calculation—his capability of getting through, without any apparent exertion, the immense transactions in which he was concerned—his coolness and judgment—enabled him to leave all his contemporaries at the Stock Exchange far behind.” His son later declared that his fathers success was based upon his observation that people in general exaggerated the importance of events. “If therefore, dealing as he dealt in stocks, there was reason for a small advance, he bought, because he was certain the unreasonable advance would enable him to realize; so when stocks were falling, he sold in the conviction that alarm and panic would produce a decline not warranted by circumstances.

It was a curiously upside-down arrangement: the theoretical stockbroker versus the practical divine—particularly curious since the theoretician was at home in

the world of money whereas the man of facts and figures was utterly at sea. ,

During the Napoleonic Wars Ricardo was an unc c - writer in a syndicate which bought government securi- ties from the Treasury and then offered them to the subscribing public. Ricardo often did Malthus a favor and carried him for a small block of securities on which the parson made a modest profit. On the eve of Waterloo Malthus thus found himself a small "bull on the Ex- change and the strain was too much for his nerves, lie wrote to Ricardo urging him "unless it is wrong or in- convenient ... to take an early opportunity of realizing a small profit on the share you have been good enough to promise me." Ricardo did, but with the stronger stay- ing power of the professional speculator bought himself into a maximum bull position. Wellington won; Ricaido made an immense killing, and poor Malthus could not help being discomfited. Ricardo, on the other hand, wrote casually to the Reverend, "This is as great an advantage as ever I expect or wish to make by a rise. I have been a considerable gainer by the loan. Now for a little of our old subject," and he plunges back into a discussion of the theoretical meaning of a rise in the price of com- modities.

Their endless debate went on, by letter and visit, until 1823. In his last letter to Malthus, Ricardo wrote: "And now, my dear Malthus, I have done. Like other dis- putants, after much discussion, we each retain our own opinions. These discussions, however, never influence our friendship; I should not like you more than I do if you agreed in opinion with me." He died that year sud- denly, at the age of fifty-one; Malthus was to go on until 1834. As for his opinion of David Ricardo: "I never loved anybody out of my own family so much."

Although Malthus and Ricardo disagreed on almost eveiything, they did not disagree about what Malthus had to say about population. For in his celebrated in 1798 Malthus seemed not only to elucidate the ques- tion once for all but also to shed a great deal of light

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somehow popuiatjon and poverty were related and a popular if apocryphai story of the day concemed an

landed16 C°fSt- °f CHllf Where °ne Juan Fernandez  
landed two goats in case he should later wish to find  
meat there. On revisiting the island he found that the  
goats had multiplied beyond reason, so he then landed  
l PfU«?u d°£S Wh° ,also multiplied and cut down the  
godts Thus, wrote the author, a Reverend James Town-  
shend a new kind of balance was restored. The weakest  
of both species were the first to pay the debt of nature;  
the most active and vigorous preserved their lives.” To  
which he added: It is the quantity of food which regu-  
lates the number of the human species.”

But while this paradigm recognized the balance that  
must be struck in nature, it still failed to draw the final  
devastating conclusions implicit in the problem. This  
was left for Malthus to do.

He began with a fascination in the sheer numerical  
possibilities contained in the idea of doubling. “. . . If  
any person will take the trouble to make the calculation,”  
he wrote, “he will see that if the necessaries of life could  
be obtained without limit, and the number of people  
could be doubled every twenty-five years, the popula-  
tion which might have been produced from a single pair  
since the Christian era, would have been sufficient, not  
only to fill the earth quite full of people, so that four  
should stand in every square yard, but to fill all the  
planets of our solar system in the same way, and not  
only them but all the planets revolving around the stars  
which are visible to the naked eye, supposing each of  
them ... to have as many planets belonging to it as our  
sun has.”

And in this appreciation of the staggering multiplica-  
tive powers of reproduction, he is quite right. One biolo-  
gist has calculated that a pair of animals, each pair pro-  
ducing ten pairs annually, would at the end of twenty  
years be responsible for 700,000,000,000,000,000,000 off-  
spring; and Havelock Ellis mentions a minute organism

which, if unimpeded in its division, would produce from one single tiny being a mass a million times larger than the sun—in thirty days.

But such examples of the prolific power of nature are meaningless in themselves. The vital question is: how great is the normal reproductive power of a human being? Malthus made the assumption that the human animal would tend to double his numbers in twenty-five years. In the light of his times this was a relatively modest assumption. It necessitated an average family of six, two of them were presumed to die before reaching the age of marriage. Turning to America, Malthus pointed out that the population there had in fact doubled itself every twenty-five years for the preceding century and a half, and that in some backwoods areas where life was freer and healthier, it was doubling every fifteen years!

But against the multiplying tendencies of the human race—and it is inconsequential to the argument whether it tended to double in twenty-five years or in fifty—Malthus opposed the obdurate fact that land, unlike people, cannot be multiplied. Land can be added to laboriously, but the rate of progress is slow and hesitant; unlike population, land does not breed. Hence while the number of mouths grows geometrically, the amount of cultivable land grows only arithmetically.

And the result, of course, is as inevitable as a proposition in logic: the number of people is bound, sooner or later, to outstrip the amount of food. “Taking the whole earth,” wrote Malthus in his *Essay*, “. . . and supposing the present population to be equal to a thousand millions, the human species would increase as the numbers 1, 2, 4, 8, 16, 32, 64, 128, 256, and subsistence as 1, 2, 3, 4, 5, 6, 7, 8, 9. In two centuries the population would be to the means of subsistence as 256 to 9; in three centuries as 4096 to 13, and in two thousand years the difference would be incalculable.”

Such a dreadful view of the future would be enough to discourage any man: “The view,” Malthus wrote, “has a melancholy hue. The troubled Reverend was driven to the conclusion that the incorrigible and irreconcilable

divergence between mouths and food could have only one result: the larger portion of mankind would forever be subjected to some kind of misery or other. For somehow the huge and ever potentially widening gap must be sealed: population, after all, cannot exist without food. Hence among the primitives such customs as infanticide—hence war, hence disease, and above all, poverty.

And if these are not enough: “Famine seems to be the last, the most dreadful resource of nature. The power of population is so superior to the power of the earth to provide subsistence . . . that premature death must in some shape or other visit the human race. The vices of mankind are active and able ministers of depopulation. . . . But should they fail in this war of extermination, sickly seasons, epidemics, pestilence, and plague advance in terrific array and sweep off their thousands and tens of thousands. Should success still be incomplete, gigantic inevitable famine stalks in the rear, and with one mighty blow, levels the population with the food of the world.”

No wonder poor Godwin complained that Malthus had converted friends of progress into reactionaries. For this is truly the doctrine of despair. Nothing, rescue mankind from the constant threat of drowning under its own weight but the frail reed of “moral restraint.” And how dependable is moral restraint against the great passion of love?

nothing can

The prospect that Malthus foresaw has come true. Recently the Foreign Policy Association wrote:

The statistics are almost beyond belief. Each day about 10,000 people in the underdeveloped countries die as a result of illnesses caused by malnutrition. Of every 20 children born in these countries, 10 are likely to perish in infancy from hunger or from the effects of an improper diet. Another seven may suffer physical or mental retardation.

This Malthusian nightmare will worsen, not improve, over the coming decades. For population in the backward

areas is growing faster than the output of food-growing at a rate<sup>o</sup>that, in the opinion of the world's food authorities portends a famine of unprecedented and unimagin-

matters stand/' writes J. J.

Spengler, an expert on the

population problem, "the prospect is definitely M\* thusian, with man sitting on a demographic time bomb

Can the bomb be defused? The date of the explosion can be delayed somewhat if we manage to increase the output of food. To feed (at present semistarvation levels) the three billion humans who will arrive on the planet before the year 2000 will require a doubling of world food output, and this may strain- capabilities to the breaking point. But even if a miracle quadrupled the output of food, we should only be deferring the famine by perhaps a generation. And meanwhile, there are other strains-housing the prospective billions will require the construction, in the next forty years, of as much shelter as has been built since the beginning of civilization.

If the time bomb is to be defused, all agree that it must be done through population control. Throughout history birth control has been practiced by the upper classes of all societies—which is one reason for the adage that the rich got richer and the poor got children. Now the problem is to introduce a knowledge of birth control among hundreds of millions of peasants, most of whom cannot read, do not trust doctors, and are suspicious of outsiders telling them how to change their ways of life.

Can it be done, before famine begins? Perhaps. The techniques exist. More important, there is evidence from a few areas (South Korea and Taiwan) where the effort has been made that peasants will accept birth control, not to please the economic planners, but to escape from the miserable cycle of birth followed by premature death. What we still do not know is whether the enormous administrative and medical effort needed to bring birth control to remote villages and urban slums can be mounted in time, and (to a lesser extent) whether religious oppo-

“„fL:dwTad?ed “UntrieS may hamp“ ,he \*-»«

It is too early to answer these questions. We only know that in our own day we are witnessing a last stage of the fearful progression that Malthus foresaw in 1798-

“r W^Ch fertility of humanity finally strips that of the soil, enforcing a ghastly limit on the quantity of human life.

Malthus did not aim his shafts at the then largely ignored continents of the East and South. His warning was directed at the Western World. There, miraculously, he was wrong. Something happened in England, on the Continent, in the United States, to retard the flood of new mouths. In 1860, sixty-three per cent of the couples married in Great Britain had families of four or more children; by 1925 only twenty per cent had more than four. Over the same span of time, the number of families with only one or two children increased from ten per cent of the total to more than half.

Why? What saved the West from the Malthusian doubling and redoubling? We do not fully understand the reasons; the laws of population growth are not yet completely clear. Birth control, of course, played a part. Originally it was called neo-Malthusianism, a name which would have made Malthus wince: he quite disapproved of the practice. But something else seems to have been equally important. The process of industrialization itself appears to have a tempering influence on the size of families. Marriages tend to take place later in advanced countries (this is the “moral restraint” on which Malthus pinned his slim hopes). The status of women is elevated from mere childbearers to active and working members of society. There are competing pleasures and desires which make a large family seem less desirable than in a simpler way of life.

To be sure, population in the United States is growing—and in recent years has been growing fairly fast. We too face a population problem, although we will experience it first in terms of crowding rather than hunger. As our numbers mount—there will be 300 million Americans

before the year 2000—space dwindles. Lines of traffic lengthen, national parks get more and more like vast picnic grounds, resources such as water get consumed dangerously fast. Yet nothing like famine threatens the United States, for advances in agricultural technology have raced out ahead of sheer numbers of mouths. Malthus did not see that the yield from cultivable land could grow much more rapidly than the area of land. Indeed, the problem in the United States is that agricultural productivity has been increasing too fast, that we have been threatened with uneatable surpluses of food.

Cannot the world follow our example? The problem is that American agriculture is productive because it is the beneficiary of enormous capital investments. The average American fanner works with a larger value of equipment than the average factory worker. It will be decades before sufficient capital can be accumulated in the backward lands. By then it will be too late.

None of this could be accurately foreseen in Malthus' day. In 1801 despite severe misgivings and rumors to the effect that this was merely a prelude to a military dictatorship, the first scientific census was undertaken in Great Britain. John Rickman, a civil servant and statistician, calculated that England's population had increased by twenty-five per cent in three decades. Although this was far from doubling, no one doubted that were it not for the disease and poverty of the masses, the population would have grown like an avalanche. No one saw the future slowing down of the birth rate; rather it seemed as if Britain was forever to face the mean poverty which sprang from an insatiably spawning humanity scrabbling for an insufficient supply of food. Poverty no longer seemed accidental, or an act of God, or the result even of human indifference. It seemed as though some malign providence had condemned the human race to eternal dolor, as if all of mankind's efforts toward self-improvement were made a farce through the stinginess of nature.

It was all very discouraging. Paley, who had urged a larger population "in preference to every other political

purpose whatsoever,” now became a convert to Malthus’ banner; Pitt, who had wanted his country enriched with more children, now withdrew his bill for higher poor relief in deference to the parsons opinions. Coleridge summed up the doleful outlook. “Finally, behold this mighty nation,” he wrote, “its rulers and wise men listening to-Paley and-to Malthus! It is mournful, mournful.”

Anyone who was not sufficiently depressed by Malthus had only to turn to David Ricardo.

At first glimpse this was not a particularly terrifying world—at least not after the Malthusian. The universe of David Ricardo, set forth in his *Principles of Political Economy* in 1817, is dry, spare, and condensed; there is none of the life, the lively detail of Adam Smith. Here is nothing but principle, abstract principle, expounded by an intellect which is focused on something more permanent than the changing flux of daily life. This is as basic, bare, unadorned and architectural as Euclid, but, unlike a set of pure geometrical propositions, this system has human overtones: it is a tragic system.

To understand that tragedy, we must take a moment to introduce the main characters in the drama. They are not, as we have said, people: they are prototypes. Nor do these prototypes, in the everyday sense of the word, live: they follow “laws of behavior.” There is none of the bustle of Adam Smith’s world here; instead we watch a kind of model puppet show in which the protean aspects of the real world have been reduced to a kind of one-dimensional caricature; this is the world stripped of everything but its economic motivations.

Whom do we meet? First, there are the workers, undifferentiated units of economic energy, whose only human aspect is a hopeless addiction to what is euphemistically called “the delights of domestic society.” Their incurable penchant for these delights results in every rise in wages promptly being met with an increase in population. The workers get their dry crust, as Alexander Baring put it, for without it they could not perpetuate themselves. But over the long run they are condemned by

their own weakness to a life at the margin of subsistence. Like Malthus, Ricardo saw only “self-restraint as a solution for the working masses and although he wished the workers well, he did not put too much faith in their powers of self-control.

Next we meet the capitalists. They are not Adam Smith’s conniving merchants. They are a gray and uniform lot, whose entire purpose on earth is to accumulate—that is, to save their profits and to reinvest them by hiring still more men to work for them; and this they do with unvarying dependability. Perhaps Ricardos training in the dispassionate world of international finance blinded him to the variety of motives besides money-making which motivated even a nineteenth-century industrialist; but for whatever reason, his capitalists are nothing but economic machines for self-aggrandizement. But the capitalists’ lot is not an easy one. For one thing, by competing among themselves, they quickly erase any undue profits which might accrue to one lucky soul who invented a new process or found an unusually profitable channel of trade. For another thing, their profits depend largely on the wages they have to pay, and as we shall see, this leads them into considerable difficulties.

But so far, save for the lack of realistic detail, it is not a world too far removed from that of Adam Smith. It was when Ricardo came to the landlord that things were different.

For Ricardo saw the landlord as a unique beneficiary in the organization of society. The worker worked, and for this he was paid a wage; the capitalist ran the show, and for this he gained a profit. But the landlord benefited from the powers of the soil, and his income—rent—was not held in line by either competition or by the power of population. In fact, he gained at everyone else’s expense.

We must pause for a moment to understand how Ricardo came to this conclusion, for his morbid outlook for society hangs on his definition of the landlord’s rent. Rent, to Ricardo, was not just the price one paid for the use of the soil, much as interest was the price of capital, and wages the price of labor. Rent was a very special

kind of return which had its origin in the demonstrable fact that not all land was equally productive.

Suppose, says Ricardo, there are two neighboring landlords. On one landlord's fields the soil is fertile, and with the labor of a hundred men and a given amount of equipment he can raise fifteen hundred bushels of grain. On the second landlord's fields, the soil is less fecund; the same men and their equipment will raise only one thousand bushels. This is merely a technical fact of nature, but it has an economic consequence: grain will be cheaper, per bushel, on the fortunate landlord's estate. Obviously since both landlords must pay out the same wages and capital costs, there will be an advantage to the man who secures five hundred more bushels than his competitor.

Now it is from this difference in costs that rent springs, according to Ricardo. For if the demand is high enough to warrant tilling the soil on the less productive farm, it will certainly be a very profitable operation to raise grain on the more productive farm. Indeed, the greater the difference between the two farms, the greater will be the differential rent. If, for example, it is just barely profitable to raise grain at a cost of \$2.00 a bushel on very bad land, then certainly a fortunate landowner whose rich soil produces grain at a cost of only 50 cents a bushel will gain a large rent indeed. For both farms will sell their grain in the same market, and the owner of the better ground will pocket the difference of \$1.50 in their costs.

All this seems innocuous enough. But now let us fit it into the world that Ricardo envisaged, and its unpleasant consequences will become quite clear.

To Ricardo the economic world was constantly tending to expand. As capitalists accumulated, they built new shops and factories. Therefore, the demand for laborers increased. This boosted wages, at least temporarily, for better pay would soon tempt the incorrigible working orders to avail themselves of those treacherous delights of domestic society and so to undo their advantage by flooding the market with still more workers. But-and

here is where the world of Ricardo turns sharply away from the hopeful prospects of Adam Smith—as population expanded, it would become necessary to push the margin of cultivation out further. More mouths would demand more grain, and more grain would de-

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more fields. And quite naturally, the new fields put into seed would not be so productive as those already in use for it would be a foolish farmer who had not already used the best soil available to him.

Thus, as the growing population caused more and more land to be put into use, the cost of grain would rise. So of course would the selling price of grain, and so too would the rents of well-situated landlords. And not only rents, but wages would rise, as well. For as grain became more expensive to produce, the laborer would have to be paid more, just to enable him to buy his dry crust and stay alive.'

And now see the tragedy. The capitalist—the man responsible for the progress of society in the first place—has been put in a double squeeze. First, the wages he has to pay are higher, since bread is dearer. Secondly, the landlords are much better off, since rents have been rising on good land, as worse and worse land has got pushed into use. And as the landlord's share in society's fruit increases, there is only one class that can get elbowed aside to make room for him—the capitalist.

What a different conclusion from Adam Smith's great pageant of progress! In Smith's world everybody gradually became better off as the division of labor increased and made the community more wealthy. In Ricardo's world, only the landlord stood to gain. The worker was forever condemned to the margin, for, poor fellow, he tended to chase after every wage rise with a flock of children and thereby to compete his wages right back down to subsistence. The capitalist, who worked and saved and invested, found that all his trouble was for nothing: his wage costs were higher, his profits smaller, and his landed opponent far richer than he. And the landlord, who did nothing but collect his rents, sat back and watched them increase.

No wonder that Ricardo fought against the Com Laws and showed the advantages of free trade which would bring cheap grain into Britain. No wonder the landlords fought tooth and nail for thirty years to keep cheap grain out of the country. And how natural that in Ricardos exposition the young industrialist class saw the theory which just fitted their needs. Were they responsible for low wages? No, since it was only the worker's own blindness which drove him to multiply his numbers. Were they responsible for the progress of society? Yes, and what did it avail them to expend their energies and save their profits for still further adventures in production? All they got for their pains was the dubious satisfaction of watching rents and money wages rise, and their own profits shrink. It was they who drove the economic machine, and the landlord, lolling in the back seat, who gained all the pleasure and reward. Indeed a sensible capitalist would really have to ask himself if the game were worth the candle.

Now who was to pop up and say that Ricardo was unfair to the landlords but Parson Malthus!

Let us remember that Malthus was not just an expert on the population question. He was first and foremost an economist, and he had, as a matter of fact, propounded the "Ricardian" theory of rent before it was taken up and refined by Ricardo himself. But Malthus did not draw the same conclusions from his theory as his friend. "Rents," said Malthus, in his *Principles of Political Economy*, which appeared three years after Ricardo's, "are the reward of present valour and wisdom, as well as of past strength and cunning. Every day lands are purchased with the fruits of industry and talent. \* In fact, Malthus adds in a footnote, "Mr. Ricardo himself is a landlord and a good example of what I mean."

It was not a very convincing rebuttal. Ricardo did not paint the landlord as a machinating figure of evil. He merely showed how the forces of economic development placed him unwittingly in a position where he stood to benefit from society's growth.

But we cannot stop here to trace all the permutations of this debate. What is important is that the dire predictions of rent envisioned by Ricardo never came to pass. For the industrialists finally did break the power of the landlords and they did finally secure the importation of cheap food. The bleak hillsides up which the wheat-fields were ominously climbing in Ricardos day were, within a few decades, returned to pasture. Equally important, the population never grew so fast as to swamp the resources of the country. For the Ricardian theory says that rent arises from the inequalities between the best land and the worst; obviously, if the population problem is under control, this difference will not develop to such a degree that rental returns assume socially alarming proportions. But consider for a moment the situation if Britain today were forced to feed her present population of fifty millions entirely from the produce of home-grown crops—suppose the old Corn Laws had never been repealed. Is there any doubt that Ricardo's picture of a landlord-dominated society would be a frightening reality? The problem of rent has become almost an academic side issue in the modern Western World. But this is not because Ricardo's analysis was faulty; we have been spared the Ricardian dilemma only because the tempo of industrial life has rescued us from the Malthusian plight: industrialism has not only given us a brake on births, but it has enormously increased our ability to raise food from the land at our disposal.

But while Malthus regarded the landlord as a valiant fellow who contributed to the wealth of nations (Ricardo said he contributed as a capitalist, making agricultural improvements, but not as a mere beneficiary from the property rights of soil), the parson found yet another cause for concern. He was worried over the possibility of what he called a "general glut"—a flood of commodities without buyers.

Such an idea is by no means foreign to us who have worried about depressions all our lives. But it appeared foolish beyond belief to Ricardo. England had had upsets in trade, but all of these appeared to be traceable

to some specific cause—a bank failure, or a burst of unwarranted speculation, or a war. And more important

Ricardo's mathematical mind, the concept could be shown <sup>^</sup>to be logically impossible. Therefore, it could never happen. '

Ricardo's proof had been discovered by a young Frenchman named J. B. Say. Say had two very simple propositions. First, he believed that the desire for commodities was infinite. The desire for food might be limited by the capacity of a man's stomach, as Adam Smith had said, but his desire for clothes, furniture, luxuries and ornaments seemed incalculably large. But not only was demand infinitely large, said Ricardo and Say, but the ability to purchase was guaranteed as well. For every good that was produced cost something—and every cost was some man's income. Whether that cost was wages, rent, or profits, its sale price accrued as someone's income. And so how could a general glut ever occur? The goods existed, the demand for them existed, and the incomes to buy them existed as well. Only pure perversity could prevent the market from finding the buyers it needed to clear its wares.

But although Ricardo accepted this as valid on the face of it, Malthus did not. It was not an easy argument to puncture, for it did seem logically watertight. But Malthus thus looked behind the process of swapping commodities for incomes, and came up with a strange idea. Was it not possible, he said, for the act of saving to make the demand for goods too small for the supply?

Again, to the modern world, this seems like a disturbingly fruitful line of inquiry. But Ricardo said it was plain and simple nonsense. "Mr. Malthus never appears to remember that to save is to spend, as surely, as what he exclusively calls spending," writes Ricardo in a reproving note. What he meant was that it was inconceivable to him that a man would bother to save his profits for any reason except to put them back into industry and earn still more profits.

This put Malthus in a quandary. Like Ricardo, he believed that saving meant spending—for industrial pur-

poses of course. But still, there seemed to be something in his argument—if only he could put his finger on it. He never could. He wrote, for example, to prove that accumulation was not quite so essential as Ricardo thought:

Many a merchant has made a large fortune although, during the acquisition of this fortune, there was perhaps hardly a single year in which he did not rather increase than diminish his expenditure on objects of luxury, enjoyment, and liberality.

To which Ricardo penciled this annihilatory comment:

True, but a brother merchant who avoided an increased expenditure on objects of luxury, enjoyment, and liberality with the same profits would get rich faster than him.

Poor Malthus! He never came off best in the exchange. His arguments were confused, and it was neither the fault of his generation that they failed to understand him nor his error that he failed to understand Ricardo. For Malthus was stumbling on a phenomenon which would not absorb the attention of economists for fifty years—the problem of booms and depressions, whereas Ricardo was entirely taken up with quite a different problem. For Malthus, the issue was the immensely important one of How Much Is There? For Ricardo it was the far more explosive question of Who Gets What? No wonder they disagreed so endlessly; they were talking about different things.

What was their contribution, when all the argument was over?

Ricardo's gift to the world was plain. Here was a world stripped to its essentials and laid open for everyone to examine: the watchworks were exposed. In its very unreality lay its strength, for not only did the bare structure of a greatly simplified world reveal the laws of rent, but

it elucidated as well vital questions of foreign trade money, taxation, and economic policy. By building a model world, Ricardo gave the powerful tool of abstraction to economics—a tool that is essential if the distraction of everyday life is to be pierced and its underlying mechanism understood. ®

Mai thus was never so successful in building an abstract world, and so his long-lasting academic contribution is smaller. But he pointed out the appalling problem of population and for that reason alone, his name is still alive. And he sensed, even if he could not explain, the problem of general depression which would occupy economists a century after his book.

The main issues with which both men wrestled are, in a sense, dead. For the Western World, at least, the population question is no longer a source of immediate concern—although it is a burning issue in the East and South—and the landlord's domination over the economy is a textbook curiosity. But between them, Malthus and Ricardo did one astonishing thing. They changed the viewpoint of their age from optimism to pessimism. No longer was it possible to view the universe of mankind as an arena in which the natural forces of society would inevitably bring about a better life for everyone. On the contrary, those natural forces which once seemed teleologically designed to bring harmony and peace into the world now seemed malevolent and menacing. If humanity did not groan under a flood of hungry mouths, it seemed that it might suffer under a flood of commodities without takers. And in either event, the outcome of a long struggle for progress would be a gloomy state where the worker just barely subsisted, where the capitalist was cheated of his efforts, and where the landlord gloated over his unearned and constantly growing spoil.

It was no small achievement for two men to convince the world that it was living in a fool's paradise. But they did; and so convincing was their proof that men sought to find a way out for society, not within the framework of the supposed natural laws, but in defiance of them. Malthus and Ricardo had shown that, left to itself, so-

cietv would proceed to a kind of barely living hell. No wonder that reformers said, if that is the case, we will put our efforts to fighting against the natural tendencies of society. If drifting will land us on the rocks we will swim against the current; and so the Utopian Socialist broke away from the faithful trust in the essential rightness of the world as it was. ,

In a sense, Malthus and Ricardo were the last of a generation which pinned its faith on reason, order, and progress. They were not apologists, nor were they defenders of an order of which they disapproved. Rather, they were impartial; they stood away from and above the social flux and with an impersonal eye plotted the direction of its current. And if what they saw was unpleasant—well, they were not to blame for that.

For these were the most scrupulously honest of men, who tracked down their ideas, regardless of where they led. Perhaps we should quote that footnote in which Malthus points out that Ricardo, the enemy of landlords, is a landlord himself:

It is somewhat singular that Mr. Ricardo, a considerable receiver of rents, should have so much underrated their national importance; while I, who have never received, nor expect to receive any, shall probably be accused of overrating their importance. Our different situations and opinions may serve at least to shew our mutual sincerity, and afford a strong presumption, that to whatever bias our minds may have been subjected in the doctrines we have laid down, it has not been that, against which perhaps it is most difficult to guard, the insensible bias of situation and interest.

After they were both gone, Sir James Mackintosh, the Scottish philosopher, paid them a wonderful tribute. “I have known,” he said, “Adam Smith slightly, Ricardo well, and Malthus intimately. Is it not something to say for a science that its three greatest masters were about the three best men I ever knew?”